P10

Practical 10: Data Analysis Using Time Series Analysis

Aim

To perform **data analysis using time series data in R programming**.

Theory

A **Time Series** is a sequence of data values recorded at **different time intervals**.

Examples:

- Stock prices recorded every day
- Monthly rainfall data
- Temperature recorded every hour

In **R programming**, time series data is stored in a **time-series object**.

The **ts() function** is used to create a time series object.

Syntax

timeseries.object.name <- ts(data, start, end, frequency)

Parameters

- **data** → vector or matrix containing values
- **start** → starting time of the series
- **end** → ending time of the series
- **frequency** → number of observations per time unit

Example:

- frequency = 12 → monthly data
- frequency = 4 → quarterly data

Algorithm / Procedure

1. Create a vector containing rainfall data.
2. Convert the data into a **time series object using ts()**.
3. Print the time series data.
4. Plot the graph of rainfall data.
5. Save the graph as a PNG file.

```
# Get the data points in the form of a vector
rainfall <- c(799,1174.8,865.1,1334.6,635.4,918.5,685.5,998.6,784.2,985,882.8,1071)

# Convert the data into a time series object
rainfall.timeseries <- ts(rainfall, start = c(2012,1), frequency = 12)

# Print the time series data
print(rainfall.timeseries)

# Create a PNG file for the chart
png(file = "rainfall.png")

# Plot the time series graph
plot(rainfall.timeseries)
```

```
# Save the chart
dev.off()
```